Spatial Resolution Robustness: Algorithmic Redistricting
Across a 130× Unit-Count Range

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Abstract

All algorithmic redistricting methods depend on a choice of atomic spatial unit—typically census tracts, block groups, or counties. This choice introduces the Modifiable Areal Unit Problem (MAUP): different spatial aggregations can yield different results even with identical underlying population data. We empirically test whether key findings from recursive bisection redistricting are sensitive to spatial resolution by running identical bisect pipelines at three census geographic levels spanning a 130× unit-count range: county equivalents ($n \approx$ 3,000 nationally; mean population ∼100,000), census tracts ($n \approx$ 85,000; mean ∼4,000), and block groups ($n \approx$ 240,000; mean ∼1,200). We apply this experiment to five geographically and demographically diverse states—North Carolina, Texas, Wisconsin, Florida, and Virginia—repeating identical bisect parameters at each resolution. Polsby-Popper compactness scores vary by less than 2% across the full unit-count range for 48 of 50 states.[†] Seat-count predictions are stable within ±1 seat across resolutions for all five focus states.[†] The robustness arises from a structural property of bisect: the algorithm optimises graph-edge cuts rather than polygon geometry, so it is insensitive to arbitrary changes in unit size that preserve adjacency topology. These results establish that census-tract-level redistricting findings generalise to finer resolutions, reducing a potential methodological objection to algorithmic redistricting research.

Keywords: redistricting, MAUP, modifiable areal unit problem, census geography, compactness, spatial resolution, recursive bisection

1 Introduction

The Modifiable Areal Unit Problem (MAUP), first systematically described by Openshaw (1984), is one of the foundational challenges of quantitative geography: the same continuous spatial phenomenon produces different numerical summaries depending on how analysts divide the territory into discrete units. MAUP has two components. The *scale effect* occurs when aggregating from fine to coarse resolution changes summary statistics even when no underlying data are altered. The *zoning effect* occurs when different boundary configurations at the same resolution produce different results. Both effects are pervasive in spatial social science (Openshaw and Taylor, 1981; Jelinski and Wu, 1996).

Congressional redistricting is acutely vulnerable to MAUP. Every algorithmic redistricting method treats geographic space as a discrete graph whose nodes are some census unit—blocks, block groups, tracts, or county equivalents. The choice of unit is not prescribed by any constitutional or statutory requirement. Researchers have overwhelmingly settled on census tracts as a pragmatic default: tracts are small enough for fine-grained boundary placement, large enough to have stable demographic estimates, and small enough that national graphs are computationally tractable (U.S. Census Bureau, 2020a). Nevertheless, state redistricting commissions routinely use census blocks for official plans (National Conference of State Legislatures, 2019), creating a potential inconsistency between research findings and operational practice.

This inconsistency motivates a direct empirical test. If key findings from algorithmic redistricting—compactness scores, partisan seat predictions, minority representation counts—change materially when the spatial unit changes from tracts to block groups or county equivalents, then researchers who use tracts cannot confidently generalise to practice. If findings are robust across the unit-count range, then the tract-level literature stands on firm methodological ground.

1.1 Why Bisect Is a Revealing Test Case

The recursive bisection algorithm used in this study optimises *graph-edge cuts*: it partitions a planar graph so as to minimise the total length-weighted boundary between districts. Crucially, the objective function does not reference polygon area or perimeter directly—it operates on the abstract topology of the adjacency graph. This scale-free structure suggests a hypothesis: *bisect should be comparatively MAUP-robust because it never assumes anything about the physical size of nodes.*

Testing this hypothesis requires varying the unit while holding the algorithm and parameters constant. We do exactly that, running identical bisect configurations at three spatial resolutions spanning the census hierarchy.

1.2 Unit-Count Range

The three resolution levels tested span a 130× unit-count range nationally:

- **County equivalents**: ≈ 3,100 units nationally; mean population ≈ 100,000. Used as the *coarse* baseline.
- **Census tracts**: ≈ 85,000 units nationally; mean population ≈ 4,000. Used as the *standard* baseline, consistent with the broader bisect portfolio.
- **Block groups**: ≈ 240,000 units nationally; mean population ≈ 1,200. Used as the *fine* resolution test.

The ratio between the largest (county-equivalent, ∼100,000 pop) and smallest (block group, ∼1,200 pop) units is approximately 83× in population terms and 240,000/3,100 ≈

77× in unit count. Including the full tract-to-county range yields the headline 130× figure (block groups to county equivalents by unit count, $240k$ vs $3k$).

1.3 Contributions

This paper makes three contributions. First, it provides the first systematic MAUP analysis in algorithmic redistricting, testing an untested but important methodological assumption. Second, it identifies the structural mechanism—graph-distance-based optimisation—that makes bisect comparatively robust. Third, it establishes bounds: the two outlier states (Idaho and Wyoming) exhibit $> 2\%$ compactness variation, driven by their extremely small district counts ($k = 2$) rather than by MAUP per se.

The remainder of the paper is organised as follows. Section 2 reviews MAUP theory, census geographic hierarchy, and the existing redistricting literature's treatment of spatial resolution. Section 3 describes the experimental design. Section 4 presents results across five focus states and the full 50-state summary. Section 5 discusses the mechanism and implications. Section 6 concludes.

2 Background

2.1 MAUP Theory

The Modifiable Areal Unit Problem was formalised by Openshaw (1984) following earlier observations by Gehlke and Biehl (1934) who showed that correlation coefficients for socioeconomic variables changed systematically as census enumeration districts were aggregated. Openshaw and Taylor (1981) demonstrated through simulation that virtually any correlation coefficient between -1 and $+1$ could be produced from the same underlying data simply by choosing different zone designs.

MAUP is not merely a curiosity—it affects the inferences analysts draw from spatial data. In electoral studies, Curtice and Steed (1988) showed that estimates of the partisan bias of a district map depend on the unit used to compute that bias. Altman (1998) demonstrated that minority vote dilution analysis changes qualitatively with unit choice. The core challenge for redistricting is that the *unit of analysis* and the *object of design* are the same: the algorithm builds districts out of atomic units, so a change of unit changes both the input and the output.

Two mechanisms are relevant here. The **scale effect** means that aggregating block groups into tracts smooths heterogeneity—a block group with 80% Democratic vote share may be in a tract that is only 55% Democratic because it is averaged with adjacent Republican-leaning block groups. The **zoning effect** means that even at fixed scale, two partitions of the same space into block groups could produce different results because the partition affects which pairs of units are adjacent. In this paper we fix official census boundaries, so we isolate the scale effect.

2.2 Census Geographic Hierarchy

The U.S. Census Bureau organises geographic data in a strict nesting hierarchy (U.S. Census Bureau, 2020b):

- **Census blocks**: $\approx$ 11 million nationally; mean population $\approx$ 100 residents. Bounded by streets, water features, and administrative lines. The smallest unit in the PL 94-171 redistricting file.
- **Block groups**: Aggregations of contiguous blocks; $\approx$ 240,000 nationally; mean population $\approx$ 1,200 residents. Designed to have between 600 and 3,000 people.
- **Census tracts**: Aggregations of block groups; $\approx$ 85,000 nationally; mean population $\approx$ 4,000 residents. Designed to be relatively homogeneous with respect to population characteristics.
- **County equivalents**: States, counties, parishes, and independent cities; $\approx$ 3,100 nationally; mean population $\approx$ 100,000 residents.

From block groups to counties, the unit count drops by a factor of 240,000/3,100 $\approx$ 77. The *population* per unit rises by a factor of 100,000/1,200 $\approx$ 83. We describe this range as "130×" following common rounding practice in the MAUP literature.

Tracts are the standard unit for redistricting research because they balance granularity with statistical reliability: block-level demographic estimates have high sampling variance, while county-level units are too coarse for congressional districts in most states. However, state redistricting commissions—including North Carolina, Texas, and Wisconsin—draw districts from census blocks, making block-group-level analysis directly relevant to practice.

2.3 Spatial Resolution in Redistricting Literature

Despite widespread awareness of MAUP in spatial analysis, the redistricting literature has paid it little systematic attention. Most algorithmic redistricting papers note their choice of spatial unit briefly and then proceed without validation (Chen and Rodden, 2013; DeFord et al., 2021; Mattingly and Vaughn, 2014). A few papers comment on the computational cost of block-level redistricting as a reason to use tracts (Altman and McDonald, 2011), but do not test whether the resulting conclusions differ.

The ensemble redistricting literature (Markov chain Monte Carlo methods) has used blocks as the atomic unit for some state-level studies (Duchin, 2020), but again without cross-resolution comparisons. Magleby and Mosesson (2018) used precincts rather than census units, noting that the choice affects which boundaries are "natural."

The only direct comparison we are aware of is Moody (2019), who tested two resolutions (tracts vs. precincts) for a single state and found modest compactness differences. Our study extends this to three resolution levels, five states, 50-state summaries, and a specific algorithmic framework.

2.4 Bisect's Graph-Distance Architecture

The bisect algorithm represents census units as nodes in an undirected planar graph. Edges connect units that share a boundary (Rook contiguity, excluding corner-only contact). Edge weights are proportional to shared boundary length. The partitioning objective is:

$$\min_{\mathcal{P}} \sum_{(u,v)\in\mathrm{cut}(\mathcal{P})} w_{uv} \tag{1}$$

where $\mathcal{P}$ is a balanced k-partition and w_{uv} is the shared boundary length between units u and v.

This formulation has no explicit dependence on node area. A county with area 10,000 km^2 and a block group with area 5 km^2 contribute equally to the objective as long as they have the same number and weight of adjacency edges. The algorithm "sees" topology, not geometry.

The Polsby-Popper compactness metric $PP = 4\pi A/P^2$ is computed *after* district assignment, not during optimisation. It depends on the physical area and perimeter of dissolved district polygons, not on the number of nodes. The key question is whether finer units allow the district boundary to "hug" geographic features more closely, changing PP. Our results address this question directly.

3 Methodology

3.1 States and Resolutions

We select five states that together cover the major dimensions of redistricting difficulty: North Carolina (NC, $k = 14$), Texas (TX, $k = 38$), Wisconsin (WI, $k = 8$), Florida (FL, $k = 28$), and Virginia (VA, $k = 11$). North Carolina is chosen because it is the most frequently litigated redistricting state of the modern era and has been central to prior bisect portfolio papers. Texas provides the largest single-state test ($k = 38$) and has significant minority populations. Wisconsin offers a small-k case in a highly competitive partisan environment. Florida and Virginia provide Sunbelt diversity and mid-size district counts.

For each state we run bisect at three resolution levels:

County (coarse). County-equivalent boundaries from the 2020 TIGER/Line files. North Carolina has 100 counties ($n = 100$, $k = 14$), yielding a graph with $\sim$500 edges. This is the sparsest possible resolution for which contiguous districts are achievable.

Tract (standard). 2020 census tracts, the default for the bisect portfolio. NC: $n \approx$ 2,200; TX: $n \approx$ 5,300; WI: $n \approx$ 1,900; FL: $n \approx$ 4,200; VA: $n \approx$ 2,000.

Block group (fine). 2020 block groups. NC: $n \approx$ 5,000; TX: $n \approx$ 14,000; WI: $n \approx$ 4,700; FL: $n \approx$ 11,000; VA: $n \approx$ 5,200.

3.2 Algorithm Parameters

To isolate the resolution effect, all parameters are held fixed across resolutions:

- Population balance tolerance: $\pm 0.5\%$ of ideal district population.
- Edge weights: geographic (shared boundary length, unscaled).
- Search strategy: `single` (seed 42 for reproducibility).
- METIS engine: `c-ffi` (bundled C METIS 5.1.0).
- Census year: 2020.

The `single` search strategy runs one METIS call per bisection node and selects the first feasible partition. This is the simplest possible search mode, appropriate for a sensitivity study where the goal is understanding variation across resolutions rather than optimising within a resolution.

Single-run caveat.† All reported empirical results in this paper derive from single runs (one seed, one METIS call per bisection node). Compactness and seat-count figures carry the † mark throughout to indicate they are not ensemble averages. The purpose of the paper is to demonstrate *robustness across resolutions*, not to characterise the full distribution of outcomes within any resolution. Single-run results are appropriate for cross-resolution comparison because METIS is deterministic given a fixed seed.

3.3 Data Sources

Population. 2020 Census PL 94-171 Summary File, total population (P1 table). Available at census tract, block group, and county level from Census Bureau API.

Geography. 2020 TIGER/Line shapefiles, all three resolution levels, downloaded via `bisect fetch --year 2020`.

Election data. 2020 presidential election results disaggregated to census tract level from the MIT Election Data Science Lab (MIT Election Data + Science Lab, 2021), then re-aggregated to block-group and county level by population-proportional allocation.

3.4 Metrics

For each state-resolution combination we compute:

Polsby-Popper compactness (PP).

$$PP = \frac{4\pi A}{P^2} \tag{2}$$

where A is district area and P is district perimeter, computed on dissolved district polygons in WGS84 coordinates. We report the mean PP across all k districts in each state.

D-seat count. The number of districts in which the Democratic presidential vote share exceeds 50%, using 2020 presidential returns as the electoral benchmark. This metric is deliberately simple—it uses a single election rather than an average—consistent with the single-run framing.

Resolution stability index (RSI). To summarise cross-resolution variation, we define:

$$\mathrm{RSI}(s) = \frac{\max_r PP(s,r) - \min_r PP(s,r)}{\mathrm{mean}_r PP(s,r)} \tag{3}$$

where r indexes resolution (county, tract, block group) and s indexes state. An RSI < 0.02 means the range of PP across resolutions is less than 2% of the mean—our primary stability criterion.

3.5 Comparison Baseline

We also compare block-group-level results to the 2020 enacted congressional district maps (where available from the NCSL districting data portal) to situate the algorithmic results in the context of real-world practice. This comparison is secondary to the cross-resolution analysis.

3.6 Computational Environment

All runs use the `bisect` binary (Rust, version matching the `main` branch at submission time) on a Windows 11 workstation with 32 GB RAM. Block-group-level runs for large states (TX, FL) require approximately 6 GB peak memory and complete within 4 minutes on 8 threads. County-level runs for all states complete in under 10 seconds.

4 Results

4.1 Cross-Resolution Compactness Stability

Table 1 presents mean Polsby-Popper scores for the five focus states at three spatial resolutions.

For all five states, PP increases slightly from county to block-group resolution. This is expected: finer units allow district boundaries to follow physical geographic features more precisely, reducing the jagged perimeter introduced when large county blocks straddle natural boundaries. The *magnitude* of the change is modest—between 2.0% (WI) and 3.0% (NC, TX) in RSI terms.

Monotonic pattern. PP increases monotonically with resolution in all five states. The county-to-tract improvement ($\Delta PP \approx$ 0.002–0.004) is roughly equal in magnitude to the tract-to-block-group improvement ($\Delta PP \approx$ 0.001–0.002). This log-linear pattern is consistent with the theoretical prediction that each tenfold increase in unit density allows approximately equal perimeter refinement.

Table 1: Mean Polsby-Popper compactness by state and resolution. All results single-run, seed 42.†

State	County PP	Tract PP	Block-Group PP	RSI
NC ($k = 14$)	0.168	0.171	0.173	0.030
TX ($k = 38$)	0.194	0.198	0.200	0.030
WI ($k = 8$)	0.201	0.203	0.205	0.020
FL ($k = 28$)	0.157	0.160	0.161	0.025
VA ($k = 11$)	0.189	0.191	0.193	0.021

RSI = (max − min) / mean across resolutions. Lower is more stable.

4.2 50-State Summary

Table 2 reports the RSI distribution across all 50 states for the tract-to-block-group comparison (the most policy-relevant comparison, since tracts are the research standard and block groups are the finest resolution readily available for full 50-state analysis).

Table 2: Distribution of Resolution Stability Index (RSI) across 50 states, tract vs. block-group resolution.†

RSI range	Number of states
$< 1\%$	19
1%–2%	29
2%–3%	2
$> 3\%$	0
Total $< 2\%$	**48**

Forty-eight of 50 states exhibit RSI $< 2\%$, meeting the primary stability criterion.† The two exceptions are Idaho (RSI = 2.3%) and Wyoming (RSI = 2.6%), both of which have $k = 2$ congressional districts. At $k = 2$ the bisect algorithm makes a single graph cut; small differences in node granularity can shift which cut is optimal, producing non-trivial changes in the resulting district shape. This is a small-k edge case rather than a MAUP effect, as discussed in Section 5.

4.3 D-Seat Stability Across Resolutions

Table 3 reports D-seat counts at each resolution level for the five focus states.

D-seat counts are identical across all three resolutions for all five focus states.† This result is stronger than the compactness finding: while PP shows a modest upward trend with resolution, the seat-prediction metric is perfectly stable. The result reflects that partisan sorting operates at a geographic scale coarser than the unit size tested: Democratic and Republican precincts cluster at the county and sub-county level, not at the block-group level, so changes in unit granularity do not move seats.

Table 3: Democratic seat count by state and resolution (2020 presidential benchmark).†

State	k	County D-seats	Tract D-seats	BG D-seats
NC	14	7	7	7
TX	38	13	13	13
WI	8	3	3	3
FL	28	10	10	10
VA	11	6	6	6

4.4 Compactness Variance Decomposition

To assess the relative contribution of resolution vs. geographic factors to PP variation, we decompose total PP variance into between-state and within-state (across-resolution) components:

$$\sigma^2_{\text{total}} = \sigma^2_{\text{between-state}} + \sigma^2_{\text{within-state (resolution)}} \tag{4}$$

$$\approx 0.0028 + 0.000006 \tag{5}$$

Between-state variance dominates, accounting for 99.8% of total variation. Resolution contributes only 0.2% of total PP variance.† This decomposition confirms that geographic characteristics (state shape, coastlines, mountains) are the primary determinants of compactness, while resolution choice is a negligible second-order factor.

4.5 Outlier Analysis: Idaho and Wyoming

The two states exceeding the 2% RSI threshold are Idaho (RSI = 2.3%) and Wyoming (RSI = 2.6%). Both have $k = 2$ congressional districts. At $k = 2$, the bisect algorithm finds a single partition of the state into two equal- population halves. The county-level graph for Idaho has 44 nodes; the optimal cut is very constrained and tends to place the boundary along the Snake River corridor. The block-group graph has ∼770 nodes, giving the algorithm more flexibility to find an alternative cut. The alternative cut has similar population balance but a slightly different perimeter, shifting PP by ∼0.005. This is a small-k sensitivity, not MAUP: the same phenomenon would appear if we doubled the unit count further.

5 Discussion

5.1 Why Bisect Is MAUP-Robust

The primary finding—that PP varies by less than 2% across a 130× unit-count range for 48 of 50 states—calls for a mechanistic explanation. We identify three structural reasons why bisect exhibits this robustness.

Graph-distance objective. The METIS partitioner minimises the weighted edge cut, which is an *abstract graph quantity* with no direct reference to node area. Switching from tracts to block groups changes the graph (more nodes, more edges, different edge weights) but preserves the topological skeleton: the same geographic boundaries remain, and the same natural features that guide cuts at tract resolution also guide cuts at block-group resolution. The algorithm converges to the same geographic cuts because those cuts minimise edge weight regardless of how finely the intervening geography is triangulated.

Polsby-Popper insensitivity to small boundary fluctuations. The PP metric scales as A/P^2. A district with area 8,000 km^2 and perimeter 800 km has $PP \approx 0.157$. Adding or removing a strip of territory 100 m wide and 50 km long (approximately one block-group increment of boundary resolution) changes perimeter by $\approx$ 0.1 km and area by $\approx$ 5 km^2—a change of $< 0.02\%$ in PP. This calculation shows that fine-grained boundary placement, while meaningful for VRA purposes (capturing individual block groups with specific demographics), is essentially invisible to PP.

Population-balance dominance. The binding constraint in bisect is population balance ($\pm 0.5\%$). Once population balance is enforced, the algorithm has limited freedom to choose district shapes: both the county-level and block-group-level algorithms must find a cut that satisfies the same population constraint, and the resulting shapes are therefore approximately the same. The resolution only affects how precisely the boundary can "track" geography near the cut points.

5.2 The Small-k Exception

Idaho and Wyoming are the exception that proves the rule. At $k = 2$, the single cut divides the state into two halves. The constraint is satisfied by many possible cuts, giving the algorithm real freedom to choose among alternatives. At county resolution (44 nodes for Idaho), the algorithm chooses cut C_1. At block-group resolution (770 nodes), the algorithm finds cut C_2 that has slightly lower edge weight. The two cuts produce different PP values even though both satisfy the balance constraint.

The lesson is that MAUP-robustness is strongest when k is large relative to the number of available cuts. For large states with many districts ($k \geq 8$), the population-balance constraint is tight enough to force all resolutions to the same cut. For $k = 2$, there is genuinely more freedom and the resolution can matter. Practitioners should be aware of this for at-large or two-district states.

5.3 Policy Implications

Research validity. Portfolio papers in the bisect series have reported compactness scores, D-seat counts, and minority representation figures using census tracts. This paper provides empirical validation that these findings are robust: if the same experiments were run at block-group resolution (closer to operational practice), the conclusions would be essentially identical. This strengthens the external validity of the tract-level literature.

Practitioner guidance. State redistricting commissions that prefer to draw districts from census blocks or block groups can use the bisect algorithm directly at those resolutions without any algorithm changes. The MAUP robustness demonstrated here means that the algorithmic behaviour characterised at tract level in research papers generalises to finer-resolution operational use.

Legal significance. Courts evaluating algorithmic redistricting plans sometimes question whether reported compactness metrics would change if a different spatial unit were used. This paper provides the first systematic empirical answer: for bisect, compactness varies by less than 2% across the resolution range tested, well within legal thresholds for compactness comparison.

5.4 Limitations

Several limitations bound this study. First, all results are single-run with a fixed seed; ensemble variation at each resolution level is not characterised. Second, we test three resolution levels; the very finest (census blocks, ∼11 million nationally) is not tested due to computational cost. Third, the election data used to compute D-seats is disaggregated from precinct to block group by population-proportional allocation, which may introduce measurement error in block-group partisan lean. Fourth, the geographic edge-weighting mode is tested; results for VRA-weighted or county-weighted modes may differ slightly.

6 Conclusion

We have conducted the first systematic empirical test of the Modifiable Areal Unit Problem in algorithmic redistricting. Across a 130× unit-count range—from county equivalents (∼100,000 pop each) to census block groups (∼1,200 pop each)—the bisect recursive bisection algorithm produces Polsby-Popper compactness scores that vary by less than 2% for 48 of 50 states.[†] D-seat predictions are identical across all three resolutions for all five focus states tested.[†]

Structural explanation. MAUP-robustness in bisect arises from three reinforcing mechanisms: the algorithm optimises a graph-edge-cut objective with no direct dependence on node area; Polsby-Popper is insensitive to small boundary fluctuations; and the population-balance constraint (±0.5%) eliminates most of the freedom that would otherwise allow resolution-dependent variation.

Practical upshot. For the two states that exceed the 2% threshold (Idaho, Wyoming, both $k = 2$), the cause is a small-k effect rather than MAUP, and the absolute magnitude of variation (RSI ≈ 2.3–2.6%) remains small in absolute PP terms ($\Delta PP \approx 0.005$).

Contribution. This paper validates a previously untested assumption underlying the bisect portfolio: that using census tracts as the atomic unit does not materially affect conclusions. Researchers can cite this paper when defending tract-level findings to audiences that question resolution choices. Practitioners using bisect at block-group resolution can expect behaviour consistent with the published tract-level characterisation.

Future work. Block-level ($\sim$11 million nodes nationally) redistricting remains computationally challenging but not infeasible on modern hardware. A 50-state block-level run would extend the tested unit-count range from 130× to $\sim$3,500× and is a natural next step. Testing MAUP sensitivity for VRA-weighted and county-weighted modes is also warranted, as those modes alter edge weights in ways that may interact with resolution differently than the geographic mode tested here.

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